

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

CODE:

```

GNU nano 6.2 sstf.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total_movement = 0;
    int visited[10] = {0};

    int i, j, min_dist, index;

    printf("Order of servicing:\n");

    for(i = 0; i < n; i++) {
        min_dist = 9999;

        for(j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);

                if(dist < min_dist) {
                    min_dist = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        printf("%d -> ", req[index]);

        total_movement += abs(head - req[index]);
        head = req[index];
    }

    printf("\n\nTotal Head Movement = %d\n", total_movement);

    return 0;
}

```

OUTPUT:

```

who@Waghmare08: ~
who@Waghmare08:~$ nano sstf.c
who@Waghmare08:~$ gcc sstf.c -o sstf
gcc: error: unrecognized command-line option '-o'
who@Waghmare08:~$ gcc sstf.c -o sstf
who@Waghmare08:~$ ./sstf
Order of servicing:
41 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 ->

Total Head Movement = 323
who@Waghmare08:~$

```